$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{2}\left(-3\,Y_1\,k^2-3\,\overset{(2)}{\kappa}_3\right) & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & Y_2\,k^2 & \frac{Y_2\,k^2}{\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_2\,k^2}{\sqrt{2}} & \frac{Y_2\,k^2}{2} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{1}{2}\left(Y_3\,k^2-2\,\overset{(2)}{\kappa}_3\right) & \frac{Y_4\,k^2+2\,\overset{(2)}{\kappa}_3}{2\,\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{Y_4\,k^2+2\,\overset{(2)}{\kappa}_3}{2\,\sqrt{2}} & \frac{1}{2}\left(-Y_5\,k^2-\overset{(2)}{\kappa}_3\right)
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{3\,k^2\,\overset{(4)}{\kappa}_{15}}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{2}{3\,\mathrm{Det}(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{1}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{-Y_5\,k^2-\overset{(2)}{\kappa}_3}{2\,\mathrm{Det}(1^-)} & \frac{-Y_4\,k^2-2\,\overset{(2)}{\kappa}_3}{2\,\sqrt{2}\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & \frac{-Y_4\,k^2-2\,\overset{(2)}{\kappa}_3}{2\,\sqrt{2}\,\mathrm{Det}(1^-)} & \frac{Y_3\,k^2-2\,\overset{(2)}{\kappa}_3}{2\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & 0 & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & \frac{2}{3\,k^2\,\overset{(4)}{\kappa}_{15}}
\end{array}
$$

Abbreviations used in matrices

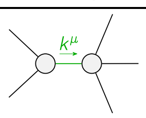
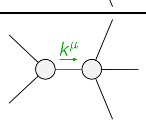
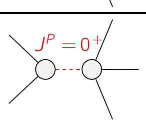
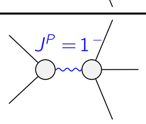
$$Y_1 == \overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2 \;\&\&\; Y_2 == 2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3 \;\&\&\; Y_3 == 3\,\overset{(4)}{\kappa}_{15}-2\,\overset{(4)}{\kappa}_2 \;\&\&\; Y_4 == 2\,\overset{(4)}{\kappa}_2+\overset{(4)}{\kappa}_7 \;\&\&$$

$$Y_5 == 3\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_2+\overset{(4)}{\kappa}_7 \;\&\&\; \mathrm{Det}(0^+) == -\frac{3}{2}\left(k^2\left(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2\right)+\overset{(2)}{\kappa}_3\right) \;\&\&\; \mathrm{Det}(1^+) == \frac{3}{2}\,k^2\left(2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3\right)$$

Lagrangian

$$
\begin{aligned}
&\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi}+\overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}+\\
&6\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-2\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}+3\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}-\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}+\\
&\overset{(4)}{\kappa}_7\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta}-\frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}+\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}+2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}^{\#1\,\alpha\beta}_{2^+} == 0$	5	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta}+3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\eta^{\alpha\beta}\,\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta}+3\left(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}+\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi}\right)$
Total # constraint(s):	8	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	3	0	$\frac{1}{2\,\overset{(4)}{\kappa}_{15}-\overset{(4)}{\kappa}_3}$
	1	0	$\frac{1}{-2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_3}$
	1	$-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2}$	$-\frac{2}{3\left(\overset{(4)}{\kappa}_1+\overset{(4)}{\kappa}_2\right)}$
	3	$\frac{6\,\overset{(4)}{\kappa}_{15}\,\overset{(2)}{\kappa}_3}{18\,\overset{(4)}{\kappa}_{15}^2-6\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+6\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_7+\overset{(4)}{\kappa}_7^2}$	$-\frac{2\left(18\,\overset{(4)}{\kappa}_{15}^2+8\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_7+\overset{(4)}{\kappa}_7^2\right)}{\overset{(4)}{\kappa}_{15}\left(18\,\overset{(4)}{\kappa}_{15}^2-6\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_2+6\,\overset{(4)}{\kappa}_{15}\,\overset{(4)}{\kappa}_7+\overset{(4)}{\kappa}_7^2\right)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		